Sampling by designing a Markov chain

Monte Carlo sampling methods using Markov chains and their applications

How can we build a chain that preserves a distribution of our choice?

W. K. Hastings · Biometrika 57(1), 97–109

Assigned reading: Sections 1, 2.1 and 2.2

Presenter A + Presenter B

We want expectations. We need samples.

$$I = \mathbb{E}_p[f(X)] = \int f(x)\, p(x)\, dx$$

The quantity

A mean, probability, or other summary of interest.

The obstacle

High-dimensional integrals can be hard to evaluate.

The opportunity

Density values may be easier to compute than samples.

Hastings changes how we obtain the samples.

Keep the average. Change the sampling mechanism.

ORDINARY MONTE CARLO

$$X_1, \ldots, X_N \overset{\text{iid}}{\sim} p$$

Draw independently from the target.

MARKOV CHAIN MONTE CARLO

$$X_0 \to X_1 \to \cdots \to X_N$$

Design transitions that preserve the target.

$$\hat{I}_N = \frac{1}{N} \sum_{t=1}^{N} f(X_t)$$

Dependent samples require care when assessing Monte Carlo error.

One target will guide the entire construction.

UNNORMALIZED WEIGHTS

$w_0 = 3 \qquad w_1 = 1$

DESIRED STATIONARY PROBABILITIES

$\pi = (3/4,\ 1/4)$



Desired long-run occupation: three times as much time in state 0.

Which transitions will produce this target?

A stationary distribution survives one transition.

$$p_{ij} = \Pr(X_{t+1} = j \mid X_t = i)$$

$$p_{ij} \geq 0 \qquad \sum_j p_{ij} = 1$$

Current distribution

π

one step using P

Next distribution

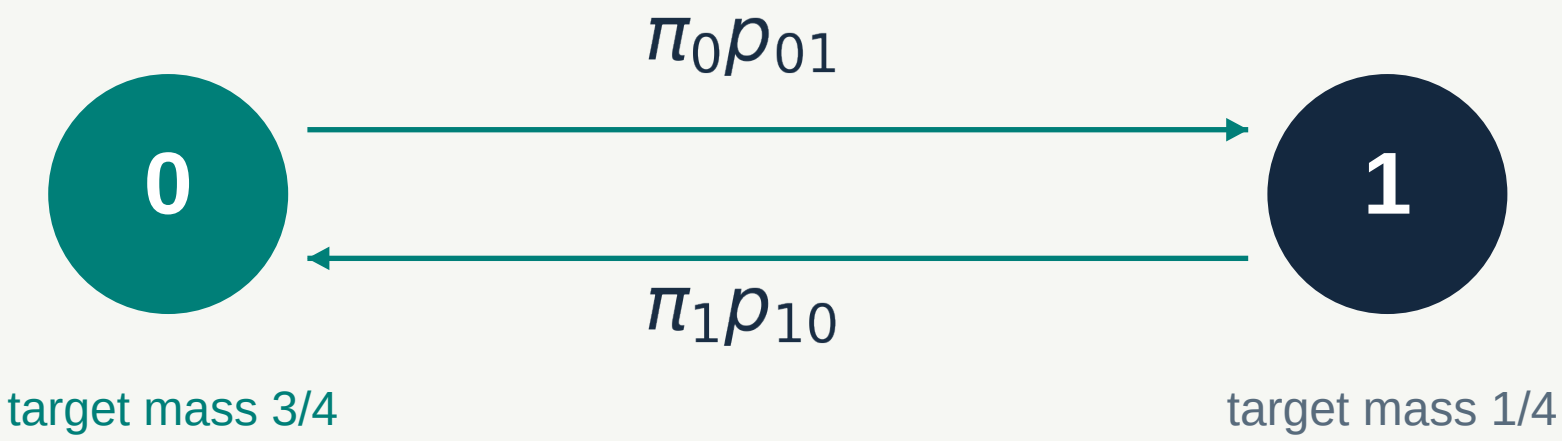
πP

$$\pi P = \pi$$

The distribution stays the same. Individual states can still change.

Balance probability flow in both directions.

$$\pi_i p_{ij} = \pi_j p_{ji}$$



Detailed balance is a sufficient condition for stationarity.

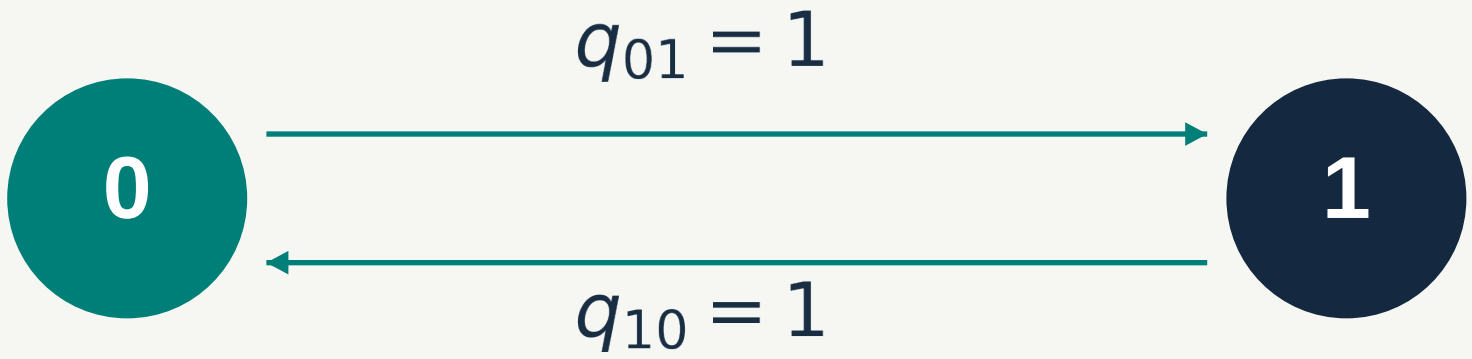
Equal weighted flows do not require equal transition probabilities.

A convenient proposal may have the wrong target.

$$Q = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Always propose the other state.

Easy to simulate. Not yet the chain we want.



Q preserves (1/2, 1/2). Our target is (3/4, 1/4).

A proposal becomes a move only if we accept it.

1

PROPOSE

Draw a candidate j from row i of Q.

2

ACCEPT OR REJECT

Accept with probability α. Otherwise remain at i.

3

RECORD

Record the next state, including repeated states.

$$p_{ij} = q_{ij}\alpha_{ij} \quad (i \neq j)$$

$$p_{ii} = 1 - \sum_{j \neq i} q_{ij}\alpha_{ij}$$

Staying can include a rejected move or a proposal of the current state.

We have the proposal. Now design the correction.

01 **Choose the target**

Done: weights 3 and 1.

02 **Choose a proposal**

Done: propose switching states.

03 **Choose acceptance probabilities**

Next: preserve the desired probability flow.

How should acceptance compensate for the proposal?

Accept according to target mass and proposal flow.

$$r_{ij} = \frac{\pi_j q_{ji}}{\pi_i q_{ij}} \qquad \alpha_{ij} = \min(1, r_{ij})$$

r ≥ 1

Accept the proposed move.

r < 1

Accept with probability r.

$$q_{ij} = q_{ji} \implies \alpha_{ij} = \min(1, \pi_j/\pi_i)$$

The reverse/forward proposal ratio corrects an asymmetric proposal.

A short correctness check: the same flow both ways.

FORWARD FLOW

$$\pi_i q_{ij} \alpha_{ij} = \min(\pi_i q_{ij}, \pi_j q_{ji})$$

REVERSE FLOW

$$\pi_j q_{ji} \alpha_{ji} = \min(\pi_j q_{ji}, \pi_i q_{ij})$$

$$\text{Detailed balance} \implies \pi P = \pi$$

A sufficient construction; stationarity does not require reversibility.

Hastings gives a family of acceptance rules.

$$\alpha_{ij} = \frac{s_{ij}}{1 + \pi_i q_{ij} / (\pi_j q_{ji})}$$

Choose symmetric s, with acceptance probabilities between 0 and 1.

Barker

$$\alpha_{ij} = \frac{r_{ij}}{1 + r_{ij}}$$

Metropolis–Hastings

$$\alpha_{ij} = \min(1, r_{ij})$$

Detailed balance does not uniquely determine the acceptance rule.

The normalizing constant disappears from the ratio.

$$\pi_i = \frac{w_i}{Z} \qquad Z = \sum_k w_k$$

$$r_{ij} = \frac{(w_j/Z)q_{ji}}{(w_i/Z)q_{ij}} = \frac{w_j q_{ji}}{w_i q_{ij}}$$

Evaluate weights and proposal probabilities.
No need to compute the potentially difficult sum or integral Z.

This removes normalization work; it does not guarantee fast exploration.

For our target, one direction needs more rejection.

Weights: 3 and 1. Proposal: always switch.

From 0 to 1

$$\alpha_{01} = \min(1, 1/3) = 1/3$$

Accept one third of proposals.
Otherwise stay in state 0.

From 1 to 0

$$\alpha_{10} = \min(1, 3) = 1$$

Accept every proposal.
Leave state 1 immediately.

State 0 gains its larger occupation through repeated visits and stays.

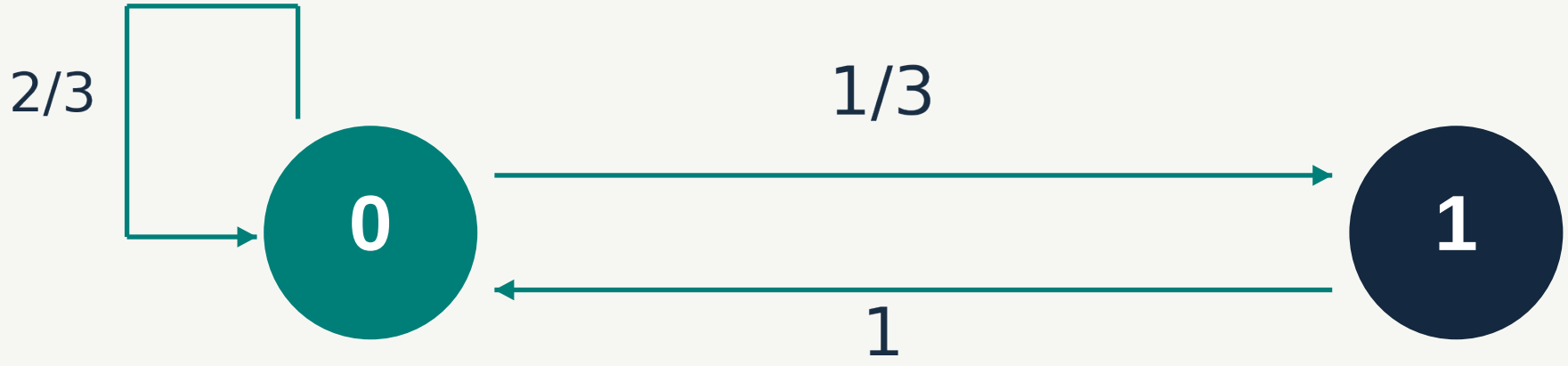
Construct P, then verify the distribution we wanted.

$$P = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ 1 & 0 \end{bmatrix}$$

$$(3/4,\ 1/4)P = (3/4,\ 1/4)$$

CHECK WEIGHTED FLOWS

$$\frac{3}{4} \cdot \frac{1}{3} = \frac{1}{4} \cdot 1 = \frac{1}{4}$$



Rejected proposals are part of the sample.

What happens if we keep only accepted moves?

ILLUSTRATIVE RECORDED TRAJECTORY · INCLUDES EVERY TIME STEP



IF REJECTED REPEATS ARE DELETED

$0 \to 1 \to 0 \to 1 \to \cdots$

1 : 1

In this example, accepted moves alternate: the target 3 : 1 is lost.

A small algorithm, with conditions we must check.

```
choose X in the target support
repeat N times:
    draw candidate J from Q[X, :]
    draw U uniformly in (0, 1)
    if U < alpha(X, J):
        X = J
    record X
```

Use the MH rule above; set α(X, X) = 1.

OUR FINITE-STATE EXAMPLE

Both states communicate.

State 0 has a self-loop.

The chain is irreducible and aperiodic.

Design the transitions. Preserve the target.

01	**Construct**	Choose a proposal and an acceptance rule.
02	**Preserve**	Balance weighted flows to obtain $\pi P = \pi$.
03	**Compute**	Use unnormalized weights; Z cancels.
04	**Record**	Keep every state, including rejected moves.

Which part of the construction makes the target distribution stationary?

An asymmetric proposal needs the correction factor.

$$Q = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ 1 & 0 \end{bmatrix}$$

$$\alpha_{01} = \min\left(1, \frac{1 \cdot 1}{3 \cdot (1/2)}\right) = \frac{2}{3}$$

$$\alpha_{10} = \min\left(1, \frac{3 \cdot (1/2)}{1 \cdot 1}\right) = 1$$

$$p_{01} = \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3} \qquad p_{10} = 1$$

The corrected proposal produces the same P as our main example.

What can the symmetric factor s actually be?

$$A = \pi_i q_{ij} > 0 \qquad B = \pi_j q_{ji} > 0 \qquad r = B/A$$

$$\alpha_{ij} = s\frac{B}{A+B} \qquad \alpha_{ji} = s\frac{A}{A+B}$$

$$0 < s \leq \frac{A+B}{\max(A,B)} = 1 + \min(r, 1/r)$$

$$s = 1 \quad \Rightarrow \quad \text{Barker}: \ \alpha = \frac{r}{1+r}$$

$$s = 1 + \min(r, 1/r) \quad \Rightarrow \quad \text{MH}: \ \alpha = \min(1, r)$$

§2.2 · Eq. (6); assume positive pairwise flows

Stationarity, convergence and independence differ.

Stationarity

$$\pi P = \pi$$

Starting in π preserves π at every time.

Marginal convergence

$$\mu P^t \longrightarrow \pi$$

For finite chains: irreducibility plus aperiodicity suffices.

Independence

$$\Pr(X_{t+1} = j \mid X_t = i) = \Pr(X_{t+1} = j)$$

Stationarity alone does not remove dependence.

Finite irreducibility is enough for long-run time averages; aperiodicity is not required for that result.

Dependence changes the uncertainty of an average.

$$\mathrm{Var}(\bar{Y}_N) = \frac{\sigma_Y^2}{N}\left[1 + 2\sum_{k=1}^{N-1}\left(1 - \frac{k}{N}\right)\rho_k\right]$$

$$Y_t = f(X_t) \qquad \rho_k = \mathrm{Corr}(Y_t, Y_{t+k})$$

Positive correlations can increase uncertainty.
Negative correlations can reduce it.

In our example, f(x) = x has lag-k correlation (−1/3)^k under stationarity.

§2.1 · pp. 98–99; stationarity and finite variance assumed

Sources and scope

Primary source

Hastings, W. K. (1970). Monte Carlo sampling methods using Markov chains and their applications. Biometrika, 57(1), 97–109.

https://doi.org/10.1093/biomet/57.1.97

Assigned scope

Sections 1, 2.1 and 2.2 (printed pp. 97–101).
The two-state construction is a teaching example, not an original example in the paper.

Course guide: construct a chain with prescribed stationarity; explain acceptance, normalization cancellation and one simple example.